

Configurar Interfaces de Roteador

Tabela de Endereçamento

Dispositivo	Interface	Endereço IP/Prefixo	Gateway padrão
R1	G0/0	172.16.20.1 /25	N/D
	G0/1	172.16.20.129 /25	N/D
	S0/0/0	209.165.200.225 /30	N/D
PC1	NIC	172.16.20.10 /25	172.16.20.1
PC2	NIC	172.16.20.138 /25	172.16.20.129
R2	G0/0	2001:db8:c0de:12::1/64	N/D
	G0/1	2001:db8:c0de:13::1/64	N/D
	S0/0/1	2001:db8:c0de:11::1/64	N/D
		fe80::2	N/A
PC3	NIC	2001:db8:c0de:12::a/64	fe80::2
PC4	NIC	2001:db8:c0de:13::a/64	fe80::2

Objetivos

- ☒ Parte 1: Configurar o endereçamento IPv4 e verificar a conectividade
- ☒ Parte 2: Configurar o endereçamento IPv6 e verificar a conectividade

Histórico

Os roteadores R1 e R2 têm, cada um, duas redes locais. Sua tarefa é configurar o endereçamento apropriado em cada dispositivo e verificar a conectividade entre as LANs.

Nota: A senha do usuário EXEC é **cisco**. A senha EXEC privilegiada é **class**.

Instruções

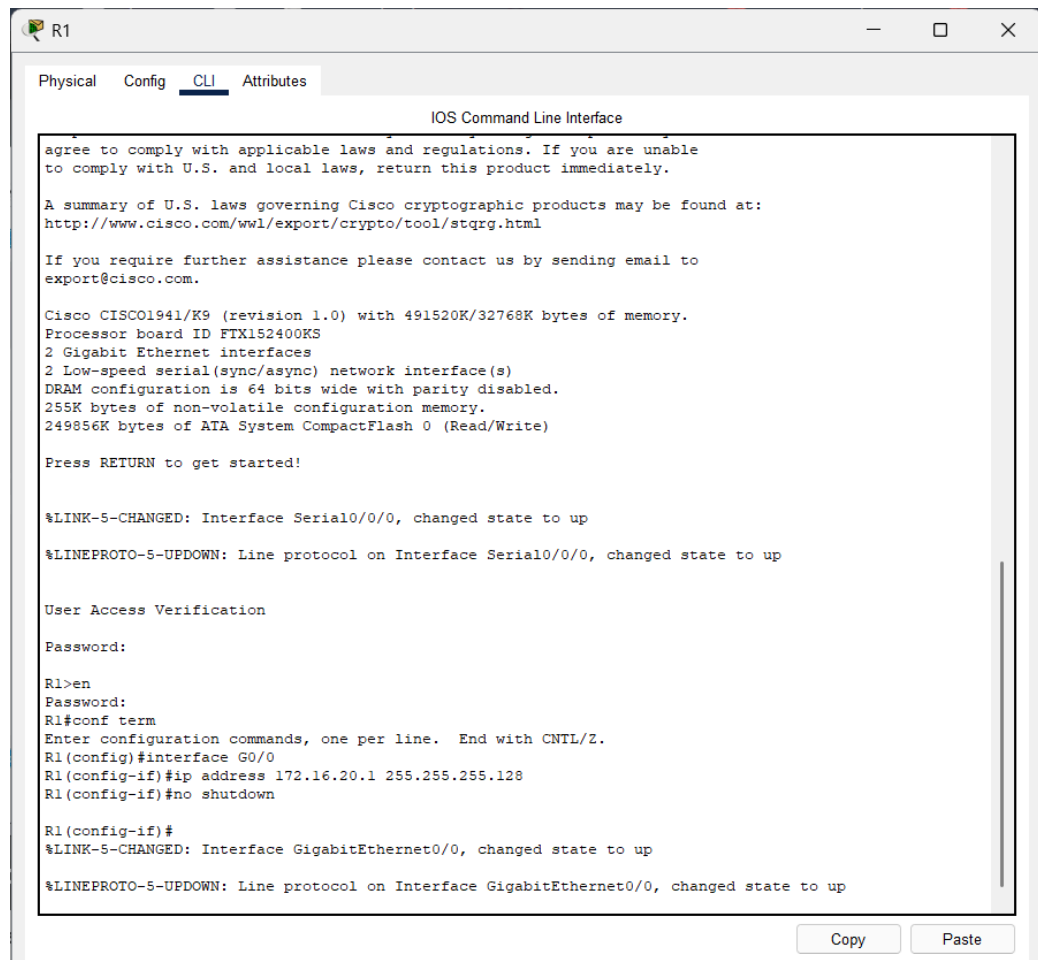
Parte 1: Configurar o endereçamento IPv4 e verificar a conectividade

Etapa 1: Atribua endereços IPv4 a R1 e aos dispositivos LAN.

Consultando a **Tabela de Endereçamento**, configure o endereçamento IP para interfaces LAN **R1**, **PC1** e **PC2**. A interface serial já foi configurada.

R:

- Cálculo da máscara
 - /25 = 255.255.255.128
- Configurando R1 G0/0



```
R1
Physical Config CLI Attributes
IOS Command Line Interface
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

User Access Verification

Password:

R1>en
Password:
R1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface G0/0
R1(config-if)#ip address 172.16.20.1 255.255.255.128
R1(config-if)#no shutdown

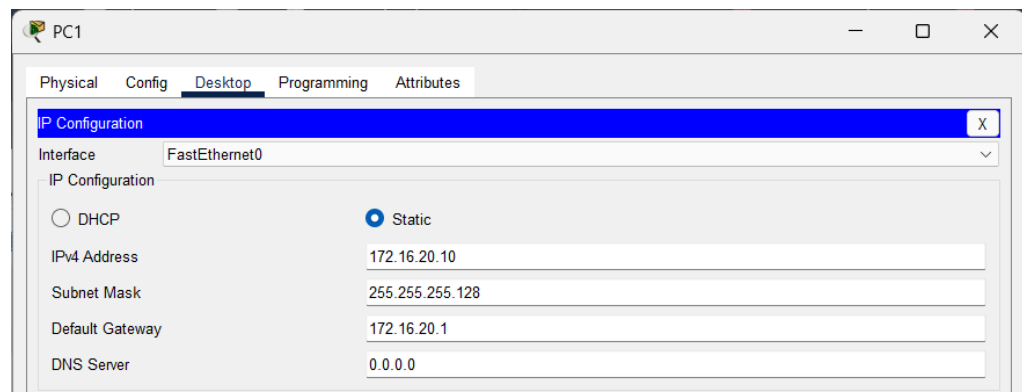
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Copy Paste
```

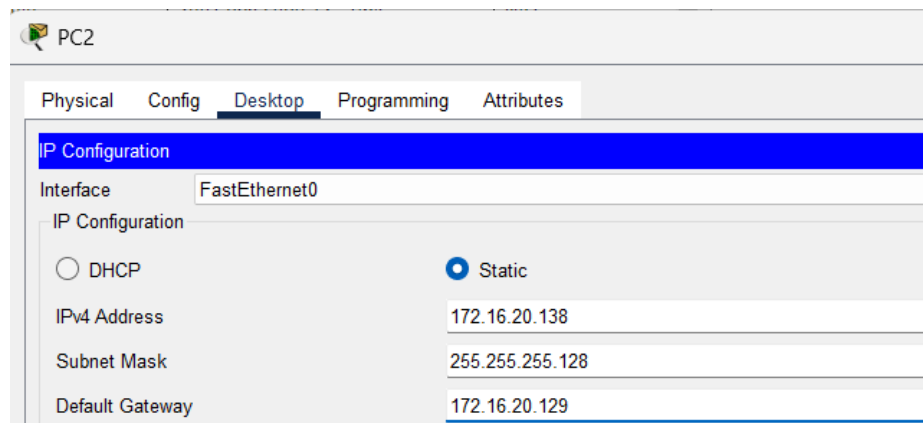
- Configurando R1 G0/1

```
R1(config-if)#  
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up  
  
R1(config-if)#exit  
R1(config)#interface G0/1  
R1(config-if)#ip address 172.16.20.129 255.255.255.128  
R1(config-if)#no shutdown  
  
R1(config-if)#  
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up  
  
R1(config-if)#
```

- Configurando PC1



- Configurando PC2

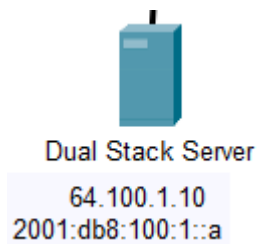


Etapa 2: Verifique a conectividade.

PC1 e PC2 devem conseguir fazer *ping* entre si e o **Servidor de Pilha Dupla**.

R:

- Dado o **Servidor de Pilha Dupla**:



- No PC1

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.20.138

Pinging 172.16.20.138 with 32 bytes of data:

Request timed out.
Reply from 172.16.20.138: bytes=32 time<1ms TTL=127
Reply from 172.16.20.138: bytes=32 time<1ms TTL=127
Reply from 172.16.20.138: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.20.138:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 64.100.1.10

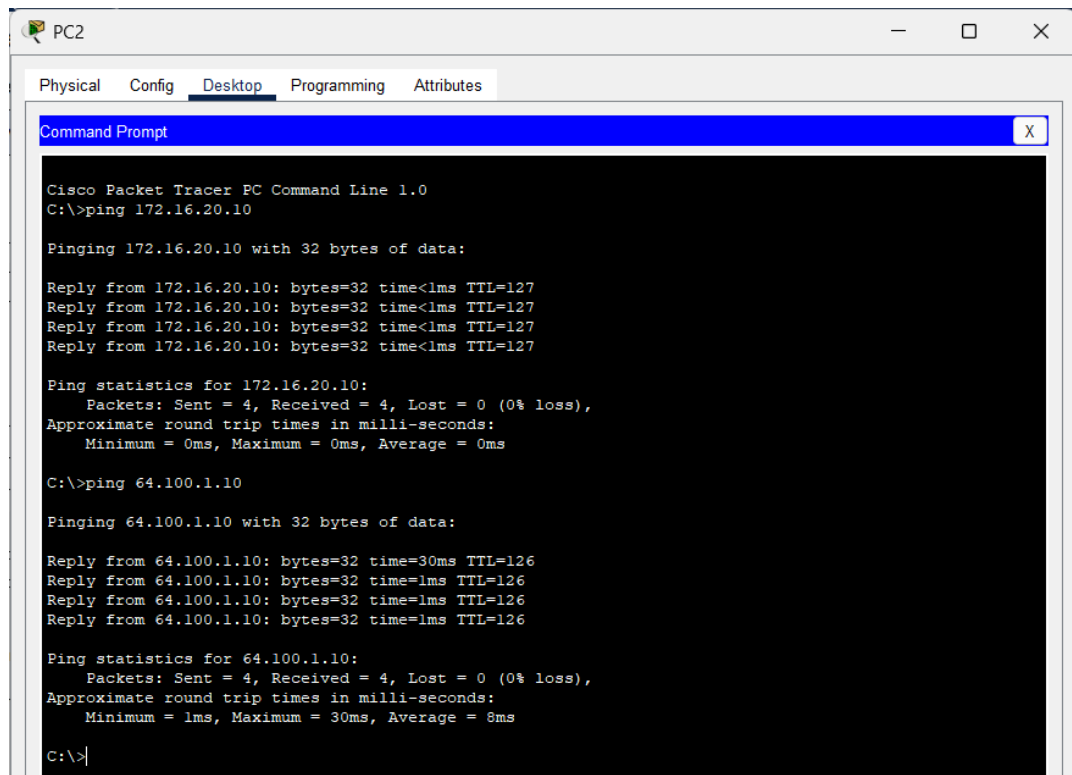
Pinging 64.100.1.10 with 32 bytes of data:

Request timed out.
Reply from 64.100.1.10: bytes=32 time=54ms TTL=126
Reply from 64.100.1.10: bytes=32 time=20ms TTL=126
Reply from 64.100.1.10: bytes=32 time=1ms TTL=126

Ping statistics for 64.100.1.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 54ms, Average = 25ms

C:\>
```

- No PC2



Parte 2: Configurar o endereçamento IPv6 e verificar a conectividade

Etapa 1: Atribua endereços IPv6 a R2 e aos dispositivos LAN.

Consultando a **Tabela de Endereçamento**, configure o endereçamento IP para interfaces LAN **R2**, **PC3** e **PC4**. A interface serial já foi configurada.

R:

- Configurando R2 G0/0

```
User Access Verification

Password:
Password:

R2>en
Password:
R2#conf term
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int g0/0
R2(config-if)#ipv6 address 2001:db8:c0de:12::1/64
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

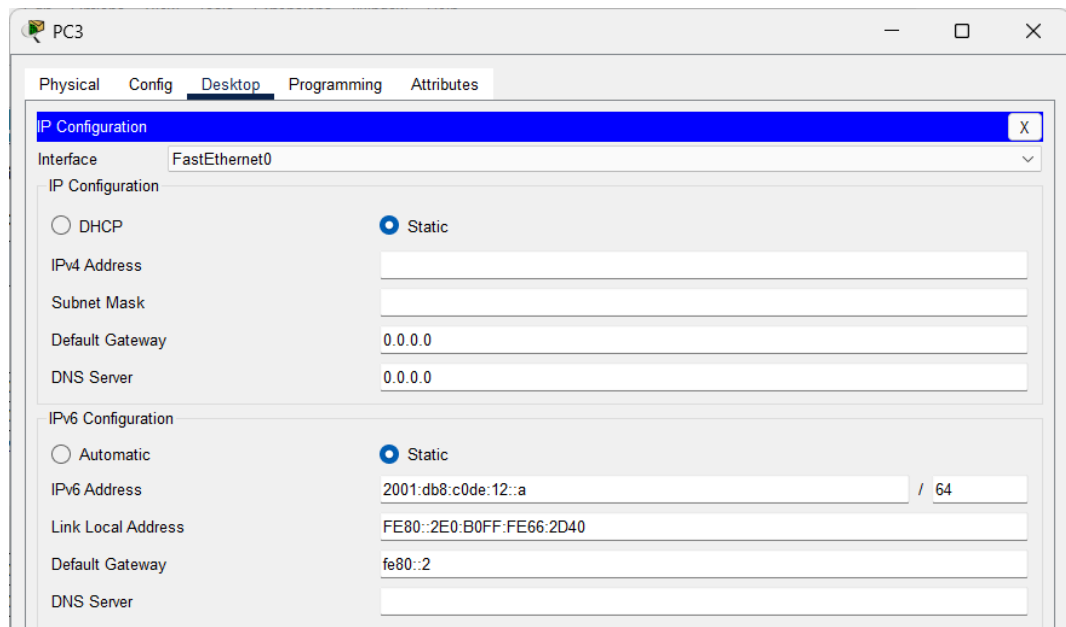
- Configurando R2 G0/1

```
R2(config)#int g0/1
R2(config-if)#ipv6 address 2001:db8:c0de:13::1/64
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

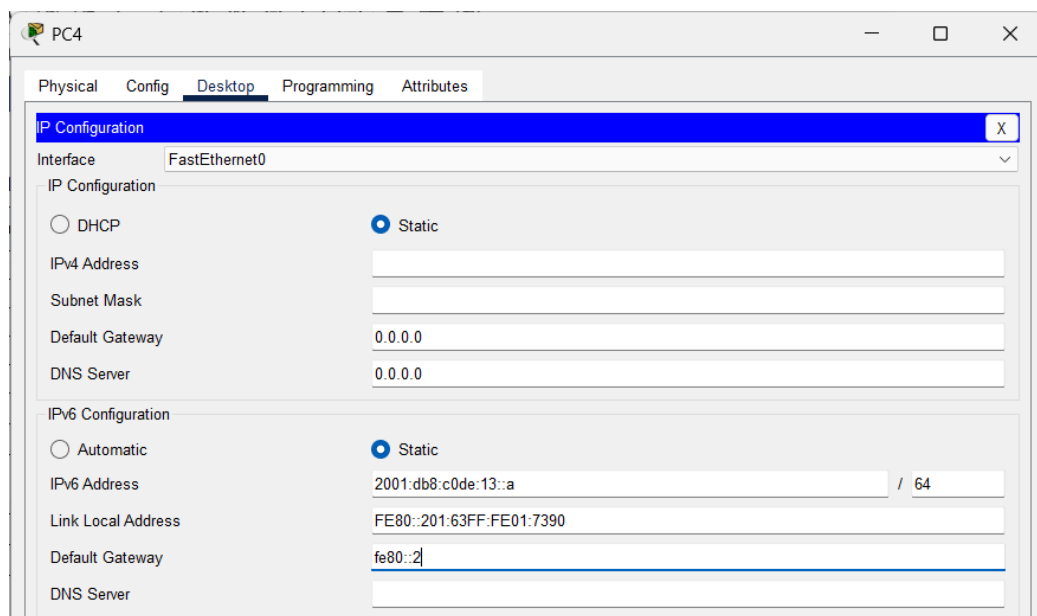
- Configurando PC3



The screenshot shows the 'PC3' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected under both 'IP Configuration' and 'IPv6 Configuration'.

Field	Value
Interface	FastEthernet0
IP Configuration	Static
IPv4 Address	
Subnet Mask	
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	Static
IPv6 Address	2001:db8:c0de:12::a / 64
Link Local Address	FE80::2E0:B0FF:FE66:2D40
Default Gateway	fe80::2
DNS Server	

- Configurando PC4



The screenshot shows the 'PC4' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected under both 'IP Configuration' and 'IPv6 Configuration'.

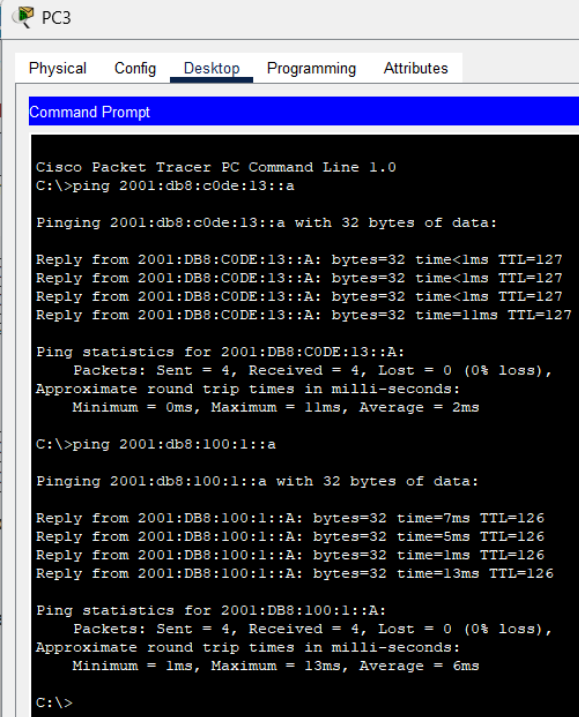
Field	Value
Interface	FastEthernet0
IP Configuration	Static
IPv4 Address	
Subnet Mask	
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	Static
IPv6 Address	2001:db8:c0de:13::a / 64
Link Local Address	FE80::201:63FF:FE01:7390
Default Gateway	fe80::2
DNS Server	

Etapa 2: Verifique a conectividade.

PC3 e PC4 deve poder executar *ping* um ao outro e o Servidor de Pilha Dupla.

R:

No PC3



```
PC3
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:c0de:13::a

Pinging 2001:db8:c0de:13::a with 32 bytes of data:

Reply from 2001:DB8:C0DE:13::A: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:C0DE:13::A: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:C0DE:13::A: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:C0DE:13::A: bytes=32 time=11ms TTL=127

Ping statistics for 2001:DB8:C0DE:13::A:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>ping 2001:db8:100:1::a

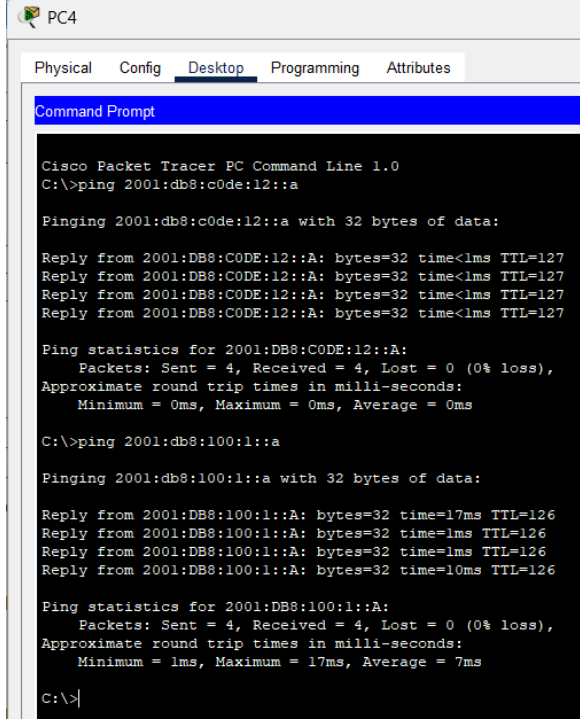
Pinging 2001:db8:100:1::a with 32 bytes of data:

Reply from 2001:DB8:100:1::A: bytes=32 time=7ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=5ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=13ms TTL=126

Ping statistics for 2001:DB8:100:1::A:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 6ms

C:\>
```

No PC4



```
PC4
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:db8:c0de:12::a

Pinging 2001:db8:c0de:12::a with 32 bytes of data:

Reply from 2001:DB8:C0DE:12::A: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:C0DE:12::A: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:C0DE:12::A: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:C0DE:12::A: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:C0DE:12::A:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:db8:100:1::a

Pinging 2001:db8:100:1::a with 32 bytes of data:

Reply from 2001:DB8:100:1::A: bytes=32 time=17ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:100:1::A: bytes=32 time=10ms TTL=126

Ping statistics for 2001:DB8:100:1::A:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 7ms

C:\>
```

Conclusão

Cisco Packet Tracer - D:\02_Areas\studies\FATEC\2025-1\IRC011 - Protocolos e Roteamento em Redes de Computadores(CCNA\1.4.7-packet-tracer---configure-router-interfaces_pt-BR.pka - Guest - 2025-07-03 19:28:43

FileEditOptionsViewToolsExtensionsWindowHelp

Activity Results

Time Elapsed: 00:44:32

Overall FeedbackAssessment ItemsConnectivity Tests

Expand/Collapse AllShow Incorrect Items

Assessment Items	Status	Points	Component(s)	Feedback
Network				
PC1				
Default Gateway	Correct	2	IPv4 Address Conf...	
Ports				
FastEthernet0				
IP Address	Correct	2	IPv4 Address Conf...	
Subnet Mask	Correct	1	IPv4 Address Conf...	
PC2				
Default Gateway	Correct	2	IPv4 Address Conf...	
Ports				
FastEthernet0				
IP Address	Correct	2	IPv4 Address Conf...	
Subnet Mask	Correct	1	IPv4 Address Conf...	
PC3				
Default Gateway IPv6	Correct	2	IPv6 Address Conf...	
Ports				
FastEthernet0				
(deprecated) IPv6 Addresses				
2001:DB8::C0DE:12::A	Correct	2	IPv6 Address Conf...	
IP Address	Correct	1	IPv6 Address Conf...	
Prefix Length	Correct	1	IPv6 Address Conf...	
PC4				
Default Gateway IPv6	Correct	2	IPv6 Address Conf...	
Ports				
FastEthernet0				
(deprecated) IPv6 Addresses				
2001:DB8::C0DE:13::A	Correct	2	IPv6 Address Conf...	
IP Address	Correct	1	IPv6 Address Conf...	
Prefix Length	Correct	1	IPv6 Address Conf...	
R1				
Ports				
GigabitEthernet0/0				
IP Address	Correct	10	IPv4 Address Conf...	
Port Status	Correct	2	IPv4 Address Conf...	
Subnet Mask	Correct	8	IPv4 Address Conf...	
GigabitEthernet0/1				
IP Address	Correct	10	IPv4 Address Conf...	
Port Status	Correct	2	IPv4 Address Conf...	
Subnet Mask	Correct	8	IPv4 Address Conf...	
R2				
Ports				
GigabitEthernet0/0				
(deprecated) IPv6 Addresses				

Score100/100

Item Count24/24

Component	Items/Total	Score
IPv4 Address Configuration	12/12	50/50
IPv6 Address Configuration	12/12	50/50

Close

Cisco Packet Tracer - D:\02_Areas\studies\FATEC\2025-1\IRC011 - Protocolos e Roteamento em Redes de Co...

FileEditOptionsViewToolsExtensionsWindowHelp

Activity Results

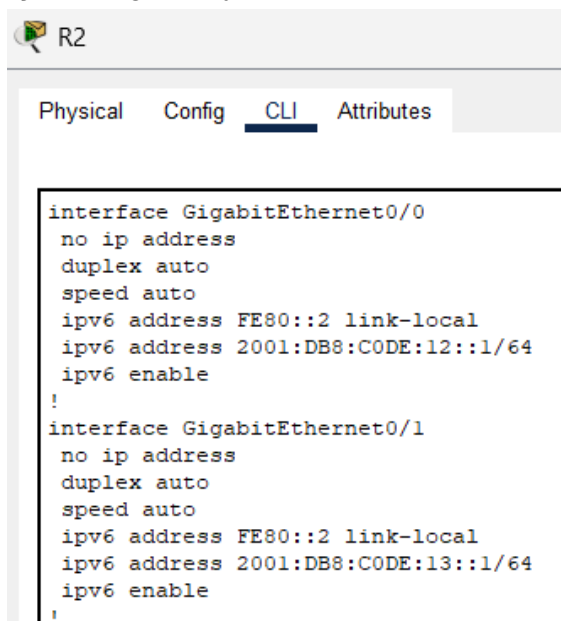
Time Elapsed: 00:45:15

Overall FeedbackAssessment ItemsConnectivity Tests

Parabéns! Você concluiu com êxito o Packet Tracer - Configure a atividade de interfaces IPv4 e IPv6

Close

Nos computadores configurados com IPv4, o default gateway foi o endereço IP das interfaces de rede (G0/0 e G0/1). Nos computadores configurados com IPv6, fez-se uso do link local para a configuração dos gateways dos computadores:



```
R2
Physical Config CLI Attributes
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:C0DE:12::1/64
ipv6 enable
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:C0DE:13::1/64
ipv6 enable
!
```